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This version: May 2, 2026
https://personal.ntu.edu.sg/nprivault/indext.html

Index

σ-algebra, 820
σ-field, 820
R code, 11, 155–157, 159, 163, 169, 171, 187–189, 213, 217, 220, 232, 233, 239, 241, 247, 248, 255, 264, 265, 267, 268, 281, 290, 344, 365–367, 373, 377, 382, 383, 392, 425, 427, 432, 434, 436, 494, 530, 588, 611–613, 615, 616, 622, 623, 634, 636, 663, 664, 682, 717, 736, 739, 742, 743, 746, 750, 761, 762, 772, 784–787, 791, 810, 811, 837, 840, 1031, 1063, 1064, 1080, 1087, 1144
R package
  bizdays, 255
  fOptions, 366, 559, 1080, 1144
  quantmod, 232, 362, 427, 494, 530, 621, 663, 762, 783
  RQuantLib, 717
  Sim.DiffProc, 233, 623
  YieldCurve, 663

absence of arbitrage, 27, 624
abstract Bayes formula, 577
accreting swap, 670
adapted process, 168
adjusted close price, 232, 763
adjustment
  convexity, 358
admissible portfolio strategy, 205
affine model, 618
affine PDE, 347, 618, 1192
American
  binary option
    finite expiration, 568
    perpetual, 568
  forward contract, 570
  option
    butterfly, 561
    call, 537, 547
    dividend, 562, 565
    finite expiration, 552
    perpetual, 537, 547
    put, 537
amortizing swap, 670
annuity
  measure, 708, 728, 730
  numéraire, 708, 726
annuity numéraire, 669
approximation
  gamma, 640
  lognormal, 490
arbitrage
  absence, 27, 624
  calendar, 385
  continuous time, 205
  discrete time, 62
  forex, 23
  maturity, 385
  opportunity, 24
  price, 15, 42, 74, 90, 286
  strike, 385
  triangular, 23
arcsine distribution, 418
arithmetic average, 479
Asian
  forward contract, 304, 509
Asian option, 479, 481
  basket, 497
  call, 479
  dividends, 512
  hedging, 511
asset pricing
  first theorem, 29

continuous time, 207, 674
discrete time, 73
second theorem, 36
continuous time, 210
discrete time, 74
at the money, 63, 98, 253, 305
ATM, 63, 98, 253, 305
attainable, 34, 40, 89, 210
automated market maker (AMM), 51, 226

Bachelier model, 192, 216, 263, 271, 299, 306, 388, 692, 1056
backward
induction, 100, 102
stochastic differential equation, 311
backward-looking bond price, 667
Bank for International Settlements, 8
Barone–Adesi & Whaley approximation, 559, 561, 1144
barrier
level, 424
barrier forward contract, 449
down-and-in
long, 449, 1091
down-and-out
long, 449, 1092
up-and-in
long, 449, 1087
up-and-out
long, 449, 1089
barrier option, 65, 421, 424
down-and-in
call, 425, 438
put, 425, 440
down-and-out
call, 425, 434, 447
put, 425, 436
down-knock-in forward, 451
hedging, 446
in-out parity, 425
up-and-in
call, 425, 439
put, 425, 441
up-and-out
call, 425, 426
put, 425, 432
up-knock-in forward, 451
basis point, 719
basket option, 300, 497
BDT model, 651
bear spread option, 301, 992
Bermudan swaption, 717
Bernoulli
distribution, 834
Bessel function, 614, 640
BGM model, 687, 702
binary
tree, 75, 135
binary option, 64, 129, 270, 308, 606, 891
American
finite expiration, 568
perpetual, 568
barrier, 450
binomial
distribution, 834
model, 75
dividends, 138
transaction costs, 135
BIS, 8
bisection method, 268, 365
bizdays (R package), 255
Black
(1976) formula, 704
LIBOR caplet formula, 702
SOFR caplet formula, 707
SOFR swaption formula, 715
swaption formula, 713
Black–Derman–Toy model, 651
Black–Scholes
calibration, 369
formula, 261, 267, 289, 592, 688
call options, 237, 703, 708
put options, 245, 246, 704
PDE, 235, 260, 268, 444, 447, 815
with jumps, 799
bond
convertible, 647, 865
convexity, 652
corporate, 647
duration, 648, 652
immunization, 1195
ladder, 708
option, 605, 699, 1176
pricing
PDE, 628, 685
yield, 637
zero-coupon, 623
Borel–Cantelli Lemma, 199, 824
boundary condition, 815
box spread option, 302
break-even
rate, 669, 670
strike price, 52
underlying asset price, 99, 253, 991
Brent, 5, 11
Bretton Woods, 574
bridge model, 690
Brownian

This version: May 2, 2026
https://personal.ntu.edu.sg/nprivault/indext.html

bridge, 194, 643
extrema, 408
motion, 149
geometric, 274
Lévy's construction, 157, 196, 964
series construction, 154, 158
BSDE, 311
bubble, 315
bull spread option, 301, 992
business time, 255
butterfly option, 303, 996
American, 561
buy back guarantee, 8, 594
buy limit, 521

calendar arbitrage, 385
calendar time, 255
call
level, 424
option, 8
price, 425, 432
spread collar option, 131
swaption, 713
call-put parity, 128, 247, 291, 310, 378, 592, 1099
callable
bear contract, 65, 424–426
bull contract, 425, 432
Cantor function, 392
Cantor set, 392
cap pricing, 705
Capital Asset Pricing Model (CAPM), 313, 1022
caplet pricing, 701
cash settlement, 7, 9, 62, 245
cash-or-nothing option, 64, 606
Category 'N' CBBC, 425
Category 'R' CBBC, 425, 448
cattle futures, 237
Cauchy
distribution, 831
sequence, 944
CBBC, 65, 424–426, 432
Category 'N', 425
Category 'R', 425, 448
rebate, 425, 448
residual, 425, 448
CBOE, 613
Volatility Index®, 378
CEV model, 317, 617
CFD, 644
change of measure, 282
change of numéraire, 576, 592
characteristic
function, 846
Chasles relation, 176
Chi square distribution, 307, 614, 1009
Chicago Board Options Exchange, 613
chooser option, 310, 1014
CIR model, 195, 264, 307, 614, 647
CKLS model, 644
Clark–Ocone formula, 113, 470
cliquet
option, 305
closing portfolio value, 58
collar option, 11
call spread, 131
costless, 13
put spread, 130
complement rule, 823
complete market, 36, 40, 285
complete space, 165, 174
completeness
continuous time, 209
discrete time, 73
compound Poisson
martingale, 775
process, 744, 779, 792
compounded yield to maturity, 652
compounding
linear, 666, 667
conditional
expectation, 66, 839, 848
probability, 824, 825
conditioning, 824
constant product automated market maker (CPAMM), 51, 226
constant repayment, 54
contingent claim, 33, 62, 73, 89
attainable, 34, 40, 210
continuous-time
limit, 123
contract for difference, 644
conversion rate, 647
convertible bond, 647, 865
convexity, 652
convexity adjustment, 358
corporate bond, 647
correction
convexity, 358
correlation
perfect, 684, 694
problem, 683
correlation option, 451
cost of carry, 588
costless collar option, 13
counterparty risk, 102
counting process, 735, 737

coupon
bond, 625
rate, 635
Courtadon model, 617, 646
Cox process, 739
Cox–Ingersoll–Ross model, 195
Cox–Ross–Rubinstein model, 75, 239
dividends, 138
transaction costs, 135
credit exposure, 102
critical price, 562
CRR model, 75, 239
dividends, 138
transaction costs, 135
cumulants, 764
Gaussian, 790
Poisson, 790
cumulative distribution function, 830
joint, 397, 833
cup & handle, 1

date
of payment, 265
of record, 265
decentralized exchange (DEX), 51, 226
deflated price, 575
Delta, 103, 106, 235, 240, 242, 248, 250, 267, 271, 297, 996, 1045
hedging, 272, 295, 599, 600
Vega hedging, 357
Delta-Vega hedging, 357
density
function, 829
marginal, 398, 833
derivatives
fixed income, 695
interest rate, 695
market, 8
differential
inequalities, 544
interest rate, 312
diffusion
elasticity, 617, 646
digital option, 64, 129, 270, 308, 606, 891
discounting, 61, 203
lemma, 214, 285, 793
discrete distribution, 834
dispersion index, 641, 738
distribution
arcsine, 418
Bernoulli, 834
binomial, 834
Cauchy, 831
discrete, 834
exponential, 831
gamma, 831
Gaussian, 830
geometric, 835
Hartman–Watson, 488
invariant, 345, 354, 611, 615
leptokurtic, 787
lognormal, 122, 220, 490, 639, 831, 988
marginal, 842
negative binomial, 835
Pascal, 835
Poisson, 835
stable, 788
stationary, 345, 354, 611, 615
uniform, 830
dividend, 83, 138, 141, 264, 304, 512, 562, 565
date of payment, 265
date of record, 265
ex-date, 265
payable date, 265
dollar value, 652
dominated convergence theorem, 540, 549, 1032
Doob-Meyer decomposition, 535
Dothan model, 618, 637
down-knock-in forward option, 451
drawdown
option, 477
process, 453, 463
drift estimation, 361
drifted Brownian motion, 279
Dupire PDE, 374
duration, 648, 652

early exercise premium, 545
ECB, 664
effective gearing, 106, 253
efficient market hypothesis, 1, 72, 995
elasticity, 254
elasticity of diffusion, 617, 646
entitlement ratio, 10, 243, 249, 369–372
entropy
contract, 359
swap, 384
equivalent probability measure, 29, 37, 73, 207, 284
Esscher transform, 807
ETF, 313, 381, 1064
Euclidean path integral, 642
Euler discretization, 809
EURIBOR, 665
European option
knock-in, 451

This version: May 2, 2026
https://personal.ntu.edu.sg/nprivault/indext.html

knock-out, 451
event, 820
ex-dividend, 265, 512
excess kurtosis, 790
exchange option, 566, 595
exchange-traded fund, 313, 381, 1064
exercise price, 6
exotic option, 65, 95, 293, 421
Asian, 479
continuous time, 421, 453, 479
discrete time, 111
lookback option, 453
expectation, 836
conditional, 839, 848
exponential
distribution, 741, 831
Lévy model, 795
Vasicek model, 194, 617, 933
extrinsic value, 98, 252

face value, 623, 652
Fano factor, 641
Fatou's lemma, 276, 525, 846, 1022
FED, 665
Feller condition, 614
filtration, 67, 151, 515
finite differences
explicit scheme, 812, 815
implicit scheme, 814, 816
first theorem of asset pricing, 29, 73, 207, 674
fixed
income, 609
derivatives, 695
leg, 669
rate, 701, 706
floating
leg, 669
rate, 701, 706
strike, 65
floorlet, 704, 720
folded normal distribution, 395
fOptions (R package), 366, 559, 1080, 1144
foreign exchange, 586
option, 590
foreign exchange option, 225
forex arbitrage, 23
formula
Lévy–Khintchine, 747
smoothing, 752
Tanaka, 196, 226, 941, 963
Taylor, 986
forward
contract, 132, 236, 268, 289, 604, 655, 689, 978, 1170
American, 570, 1162
Asian, 304, 509
non-deliverable, 237
measure, 648, 696
price, 575
range, 128, 268
rate, 655, 689
agreement, 655, 689
spot, 655, 656, 658, 701, 706
swap, 668
start option, 305
swap rate, 668
forward measure, 576
forward swap
measure, 708, 728, 730
four-way collar option, 11
Fourier
synthesis, 158
transform, 347
inversion, 347
FRA, 655, 689
Fubini theorem, 754
fugazi (the), 364
futures contract, 237, 644, 897
FX option, 225

gains process, 94
Galton board, 121
Gamma
Greek, 242, 250, 1045
process, 761
gamma
approximation, 640
distribution, 831
function, 831
swap, 359
Gamma hedging, 272
gap, 783
Garman-Kohlagen formula, 590
Gaussian
cumulative distribution function, 125, 700
distribution, 238, 830
random variable, 847
gearing, 98, 253
effective, 106, 253
Geman–Yor method, 490
generating function, 195, 847
geometric
average, 482, 510
Brownian motion, 216, 274
distribution, 835

geometric mean market maker model, 51, 226
Girsanov Theorem, 282, 313, 582
  jump processes, 769, 793
Goldman–Sosin–Gatto formula, 409, 414, 453, 463
Greeks, 250, 1045
  Delta, 235, 240, 242, 248, 250, 267, 271, 297, 996, 1045
  Gamma, 242, 250, 1045
  Rho, 250, 1045
  Rhod, 271
  Theta, 250, 309, 1014, 1045
  Vanna, 1044
  Vega, 250, 271, 450, 1045, 1095
gross market value, 8
gross world product, 8
guarantee
  buy back, 8, 594
  price lock, 10
GWP, 8

Hartman–Watson distribution, 488
Hawaiian option, 480
heat
  equation, 255, 812
  map, 400
hedge and forget, 236, 896, 1171, 1256
hedge ratio, 107, 254
hedging, 35, 101, 102, 111, 292
  change of numéraire, 597
  Delta, 272, 357
  Gamma, 272
  gamma, 272
  mean-variance, 801
  quantile, 317
  static, 236, 896, 1171, 1256
  strategy, 294
  Vega, 348, 357
  Velta, 357
  with jumps, 801
Heston model, 330, 356
hexanomial model, 920
HIBOR, 665
historical
  probability measure, 280
  volatility, 331, 361
hitting
  time, 521
HJM
  condition, 674
  model, 672
  SOFR model, 678
Ho–Lee model, 618
Hull–White model, 618, 674

immunization, 1195
implied
  probability, 17
  volatility, 364
in the money, 63, 98, 253, 371, 885
in-out parity, 425, 1099
inclusion-exclusion principle, 823
independence, 824, 826, 829, 833, 835, 840, 847, 852
independent increments, 274, 772
indicator function, 827
infimum, 835
infinitesimal, 177
information flow, 68
instantaneous forward rate, 658
interest rate
  derivatives, 695
  differential, 312
  model
    affine, 618
    Constant Elasticity of Variance, 617
    Courtadon, 617, 646
    Cox–Ingersoll–Ross, 264, 307, 614
    Dothan, 618, 637
    exponential Vasicek, 194, 617, 933
    Ho–Lee, 618
    Hull–White, 618, 674
    Marsh–Rosenfeld, 617, 646
    Vasicek, 610
  short term, 609
intrinsic value, 43, 98, 252
invariant distribution, 345, 354, 611, 615
inverse Gaussian process, 762
IPython notebook, 15, 75, 96, 100, 103, 108, 110, 137, 145, 157, 158, 239, 268, 365, 369, 392, 508, 561, 613, 615, 680, 907, 919
Itô
  formula, 179, 307
    pathwise, 756
    with jumps, 757
  isometry, 161, 164, 172, 753
  process, 179, 181, 234, 981
  stochastic integral, 160, 170, 172, 273
  table, 182, 457
    with jumps, 758, 1244
ITM, 63, 98, 253, 371, 885

Jamshidian's trick, 720
Jensen's inequality, 132, 276, 483, 518, 896, 990, 1124
joint

This version: May 2, 2026
https://personal.ntu.edu.sg/nprivault/indext.html

cumulative distribution function, 397, 833
probability density function, 832
jump-diffusion process, 783

kimchi, 23
knock-in option, 451
knock-out option, 65, 425, 451
Kullback-Leibler entropy, 794
kurtosis, 787, 790

Lévy
construction of Brownian motion, 157, 196, 964
process, 761
Lévy characterization, 193
Lévy–Khintchine formula, 747
Lagrangian, 641
Laplace transform, 490, 534
law
of total expectation, 842
of total probability, 823, 826, 842
least square Monte Carlo, 558
least square regression, 619
leg
fixed, 669
floating, 669
Leibniz integral rule, 675
lemma
Neyman–Pearson, 318
leptokurtic distribution, 787
leverage, 224, 314, 320, 1022, 1039
liability, 14
LIBOR
model, 665
rate, 665
swap rate, 669, 711, 713
LIBOR-SOFR swap, 716
Lipschitz function, 595
liquidity
pool, 51, 226
provider, 51, 226
local
time, 196
volatility, 372, 815
local martingale, 315
log
contract, 269, 305, 359
option, 321, 333
return, 362
dynamics, 217, 358, 795
variance, 122, 220, 328
log variance, 220
lognormal
approximation, 490
distribution, 122, 220, 639, 831, 988
long box spread option, 302, 994
long forward contract, 805, 806
lookback option
call, 463
put, 453, 456
LOTUS, 837
LSM, 558

Macaulay duration, 652
Mandatory Call Event, 448
marginal
density, 398, 833
distribution, 842
Margrabe formula, 596
mark to market, 42, 74, 90, 237, 286, 897
market
bubble, 315
completeness, 36, 40, 73
efficiency, 995
making, 42
price of risk, 278, 284, 351, 628
volatility, 351
market terms and data, 98, 250
Markov property, 186, 595, 599
Marsh–Rosenfeld model, 617, 646
martingale, 66, 152, 273, 516
compound Poisson, 775
continuous time, 151, 206
discrete time, 69
local, 315
measure
continuous time, 205, 792
discrete time, 72
method, 285
Poisson, 772
submartingale, 516
supermartingale, 516
transform, 70, 94, 275, 523
maturity, 6
transformation, 656
maturity arbitrage, 385
maximum of Brownian motion, 389
MCE, 448
mean
hitting time, 532
reversion, 609
mean-square distance, 851
measurability, 168
Merton model, 796
method
bisection, 365
Newton–Raphson, 365

Milstein discretization, 810
Minkowski inequality, 165
model
  affine, 618
  Bachelier, 216, 263, 271, 299, 306, 388, 1056
  Barone–Adesi & Whaley, 561
  BDT, 651
  binomial, 75
    dividends, 138
    transaction costs, 135
  CEV, 317, 617
  CIR, 614, 647
  CKLS, 644
  Courtadon, 617, 646
  Dothan, 618, 637
  exponential Lévy, 795
  exponential Vasicek, 617
  hexanomial, 920
  Ho–Lee, 618
  Hull–White, 618, 674
  Marsh–Rosenfeld, 617, 646
  Merton, 796
  pentanomial, 920
  trinomial, 82, 126, 143
  Vasicek, 610
modified
  Bessel function, 614, 640
  duration, 652
moment
  generating function, 847, 1248
moment matching, 490
moments
  Gaussian, 790
  Poisson, 790
moneyness, 63, 98, 253
moving average, 480
MPoR, 278, 284, 351, 628
Musiela notation, 673

natural logarithm, 238
negative
  binomial distribution, 835
  inverse Gaussian process, 762
  premium, 29
  risk premium, 206
Nelson–Siegel, 679, 682
Newton–Raphson method, 365
Neyman–Pearson Lemma, 318
nominal value, 652
non-deliverable forward contract, 237
noncentral Chi square, 307, 614, 1009
nonlocal operator, 801
notional, 670
  principal, 719, 1219, 1220
notional amount, 8
numéraire, 206, 573
  annuity, 669, 708
  invariance, 597
numéraire invariance, 599

OLS, 619
opening
  jump, 783
  portfolio price, 58
optimal stopping, 552
option
  Asian, 479
    basket, 497
    call, 479
  at the money, 305
  barrier, 65, 421
  basket, 300, 497
  bear spread, 301, 992
  binary, 64, 129, 606, 891
  box spread, 302
  bull spread, 301, 992
  butterfly, 303, 996
  cash-or-nothing, 64, 606
  chooser, 310, 1014
  cliquet, 305
  digital, 64, 129, 606, 891
  drawdown, 477
  effective gearing, 106, 253
  exotic, 65, 95, 111, 293, 421, 453, 479
  extrinsic value, 98, 252
  foreign exchange, 225
  forward start, 305
  gearing, 98, 253
  Hawaiian, 480
  intrinsic value, 98, 252
  issuer, 15, 35
  knock-in, 451
  knock-out, 65, 425
  long box spread, 302, 994
  lookback, 453
  on average, 64, 299, 479, 510
  on extrema, 422
  out of the money, 309
  path-dependent, 111, 293
  power, 132, 216, 266, 306, 604, 897
  premium, 35, 99, 253
  ratchet, 305
  straddle, 1018
  tunnel, 126, 128
  vanilla, 235
  variance call, 336
  variance swap, 332

This version: May 2, 2026
https://personal.ntu.edu.sg/nprivault/indext.html

volatility swap, 341, 387, 1061
writer, 15, 35
zero-collar, 13
optional
sampling, 522
stopping, 522
order book, 966
Ornstein–Uhlenbeck process, 610
OTM, 63, 98, 253, 309
out of the money, 63, 98, 253, 309

Paley–Wiener series, 158
par value, 623, 652
parity
call-put, 128, 247, 291, 310, 378, 592, 1099
in-out, 425, 1099
Partial integro-differential equation, 799
partition, 826, 849
Pascal distribution, 835
path
freezing, 721
integral, 98, 422, 580, 641
Euclidean, 642
path-dependent option, 111, 293
pathwise Itô formula, 756
payable date, 265
payer
swap, 669
swaption, 710
payoff function, 7, 9, 421
PDE
affine, 347, 618, 1192
Black–Scholes, 235, 260
Heston, 345
integro-differential, 799
variational, 556
pentanomial model, 920
perfect correlation, 684, 694
Perron–Frobenius theorem, 84
physical delivery, 7, 9, 62, 245
PIDE, 799
Planck constant, 641
Poisson
compound martingale, 744, 792
distribution, 835
process, 735
compound, 779
portfolio, 22
process, 94
replicating, 103, 108
strategy, 34, 56, 89, 207, 210
admissible, 205, 210
update, 208, 211
value, 60, 90
power option, 132, 216, 266, 306, 604, 897
predictable process, 70, 93, 752
premium
early exercise, 545
kimchi, 23
negative, 29
option, 99, 253
risk, 29, 206, 278
price
critical, 562
graph, 7, 9, 12, 130, 131, 893, 894
price lock guarantee, 10
pricing, 89, 95
with jumps, 793
principal amount, 719
probability
conditional, 824, 825
density function, 829
joint, 832
distribution, 828
measure, 823
equivalent, 29, 37, 73, 207, 284
sample space, 819
space, 824
process
counting, 735
Cox, 739
drawdown, 453, 463
gamma, 761
inverse Gaussian, 761
Lévy, 761
predictable, 70, 93, 752
stable, 761
stopped, 522
variance gamma, 761
pushforward measure, 775
put
option, 5
spread collar option, 130
swaption, 715
Python code, 15, 75, 96, 100, 103, 108, 110, 137, 145, 146, 157, 158, 239, 268, 365, 369, 392, 508, 561, 613, 615, 680, 736, 739, 742–744, 746, 751, 837, 840, 907, 919
Python package
yfinance, 369

quantile hedging, 317
Quantlib, 717
quantmod (R package), 232, 362, 427, 494, 530, 621, 663, 762, 783

Radon–Nikodym, 280
Radon–Nikodym density, 576
random
  product, 844
  sum, 843
  variable, 827
range forward, 128, 268
ratchet
  option, 305
rate
  forward, 655, 689
  forward swap, 668
  instantaneous forward, 658
  LIBOR, 665, 670
    swap, 669
  LIBOR swap, 711, 713
  SOFR swap, 715
  swap, 668
realized variance, 331, 362
  swap, 332
receiver swaption, 715
recovery theorem, 48, 84
reflection principle, 421
relative entropy, 794
renewal processes, 743
replicating portfolio, 103, 108
replication, 35
repo market, 667
repurchase agreement, 667
return
  log, 362
Rho, 250, 271, 1045
Riccati equation, 633, 1197
risk
  counterparty, 102
  market price, 278, 284, 351, 628
  premium, 29, 206
risk premium, 278
risk-neutral
  measure, 16, 28, 792
    continuous time, 205, 278
    discrete time, 72
  probabilities, 16
riskless asset, 125, 213
Robinhood incident, 995
RQuantLib, 717
running maximum, 389, 390

SABR model, 356
second theorem of asset pricing, 36, 74, 210
self-financing portfolio
  change of numéraire, 599
  continuous time, 207, 209, 210, 212, 801
  discrete time, 58, 90
sell stop, 521
seller swap, 669
share right, 28
Sharpe ratio, 284
SHIBOR, 665
short rate, 609
short selling, 40, 107, 244
  ratio, 23
SIBOR, 665
Sim.DiffProc, 233, 623
singular measure, 457
skewness, 787, 789
slow-fast system, 352
smile, 366
smooth pasting, 541
smoothing formula, 752
SOFR, 667
  Black caplet formula, 707
  caplet, 706
  forward rate, 668
  HJM model, 678
  swap rate, 672, 715
  swaption, 715
spline function, 377
spot forward rate, 655, 656, 658, 701, 706
spread option, 692
square-integrable
  functions, 160
  random variables, 164
St. Petersburg paradox, 838
stability warrant, 450
stable
  distribution, 788
  process, 762
static hedging, 236, 896, 1171, 1256
stationary distribution, 345, 354, 611, 615
stochastic
  calculus, 176
  differential equations, 186
  integral, 92, 158, 168
    with jumps, 751
  integral decomposition, 113, 192, 292, 295
  process, 56
stochastic portfolio theory, 222
stop-loss/start-gain strategy, 225
stopped process, 522
stopping time, 520
  theorem, 522
straddle option, 1018
Stratonovich integral, 949
strike arbitrage, 385
strike price, 6, 34

This version: May 2, 2026
https://personal.ntu.edu.sg/nprivault/indext.html

floating, 65
string model, 694
strong Markov property, 741
submartingale, 516
super-hedging, 35, 74
supermartingale, 516
Svensson parametrization, 679
swap, 668
amortizing, 670
entropy, 384
forward measure, 708, 728, 730
gamma, 359
payer, 669
rate, 668, 670
seller, 669
variance, 332
swaption, 710
Bermudan , 717

Tanaka formula, 196, 226, 941, 963
Taylor's formula, 177, 986
telescoping sum, 671
tenor structure, 575, 668, 695
terms and data, 98, 250
ternary tree, 82, 126, 143
theorem
asset pricing, 29, 36, 73, 74, 207, 210, 674
dominated convergence, 540, 549
Fubini, 754
Girsanov, 282, 313, 582, 769, 793
Perron–Frobenius, 84
recovery, 48, 84
stopping time, 522
Theta, 250, 309, 1014, 1045
TIBOR, 665
time
business, 255
splitting, 225, 307, 962
tower property, 69, 71, 93, 100, 173, 274, 276, 295, 298, 577, 627, 842, 846, 852, 876, 919, 1204
transaction cost, 135
transform
Esscher, 807
Fourier, 347
Laplace, 490, 534
martingale, 70, 94, 523
treasury note, 613
tree
binary, 75, 135
ternary, 82, 126, 143
trend estimation, 361
triangle inequality, 165
triangular arbitrage, 23
trinomial model, 82, 126, 143
tunnel option, 126, 128
turbo warrant, 65, 424–426, 432
two-asset correlation option, 451
two-factor model, 685

uniform distribution, 830
Uniswap, 51, 226
up-knock-in forward option, 451

valuation period, 448
vanilla option, 63, 95, 235
vanna, 1044
variable rate, 701, 706
variance, 843
call option, 336
gamma process, 761
realized, 331, 362
swap, 332
variational PDE, 544, 556
Vasicek model, 610
Vega, 250, 271, 450, 1045, 1095
notional, 332
Vega hedging, 348, 357
VIX®, 378
volatility
historical, 331, 361
implied, 364
level, 332, 384
local, 372, 815
smile, 366
surface, 369
swap, 341, 387, 1061
variance swap, 332

warrant, 10, 83, 243
stability, 450
terms and data, 255
turbo, 65, 424–426, 432
weak solution, 189
West Texas Intermediate (WTI), 11
Wiener space, 2

yfinance (Python package), 369
yield, 655, 658, 701, 706
bond, 637
compounded to maturity, 652
curve, 656
data, 663
inversion, 665
YieldCurve (R package), 663

zero measure, 392

zero-
collar option, 13
coupon bond, 623

Author index

Achdou, Y. 377
Ackerer, D. 377
Albanese, C. 330, 614
Albrecher, H. 348
Allegretto, J. 559
Applebaum, D. 761
Aristotle 5
Armstrong, J. 272
Aste, T. 229
Attari, M. 348

Bachelier, L. 2, 158
Barone-Adesi, G. 559, 561
Barrieu, P. 488
Bartlett, B. 357, 1044
Benth, F.E. 497
Bergomi, L. 1051
Bermin, H. 470
Bickersteth, M. 668
Billingsley, R.S. 995
Björk, T. 44, 682
Black, F. 3, 210, 231, 651, 702, 707, 713
Borodin, A.N. 401
Bosq, D. 737
Boulding, K.E. 213, 574
Boyle, P.P. 907
Brace, A. 4, 687
Breeden, D.T. 373
Brémaud, P. 752
Brigo, D. 341, 630, 686, 1193
Briola, A. 229
Broadie, M. 358, 565
Brody, D.C. 645
Brown, C. 511
Brown, R. 2, 152
Buchen, P. 1106
Burdzy, K. 390
Buterin, V. 51, 226

Carmona, R. 307, 692
Carr, P. 334, 359, 489, 490
Chan, C.M. 65, 425
Chan, K.C. 644
Chance, D.M. 995
Charlier, C.V.L. 790
Charpentier, A. 663
Chataigner, M. 377
Christensen, B.J. 682
Christensen, J.H.E. 682
Çınlar, E. 829
Coffey, B.K. 9
Cont, R. 761, 768, 774, 783, 805
Courtadon, G. 617, 646
Cousin, A. 377
Cox, A.M.G. 315
Cox, J.C. 75, 264, 307, 614
Cramér, H. 790
Crépey, S. 480
Curran, M. 495

Da Fonseca, J. 355
Dahl, L. O. 497
Dana, R.A. 463
Dash, J. 580
Dassios, A. 477
Deelstra, G. 497
Demeterfi, K. 359
Denson, N. 604
Derman, E. 359, 374, 651
Detemple, J. 565
Devore, J.L. 819
Di Nunno, G. 111, 293, 805
Diallo, I. 497
Diebold, F.X. 682
Ding, Y. 668
Dixon, M. 377
Doob, J.L. 516, 522, 535
Dothan, L.U. 618, 637
Downes, A. 604
Dudley, R.M. 165
Dufresne, D. 487, 490
Dupire, B. 374
Durrleman, V. 307, 692
Dvoretzky, A. 390

Einstein, A. 2
El Karoui, N. 690
El Khatib, Y. 470, 473
Elliott, R.J. 545, 556, 559
Erdos, P. 390
Eriksson, J. 65, 425

Ewald, C.-O. 511

Faff, R. 1185
Feller, W. 330, 614, 1009
Fernholz, E.R. 222, 955
Filipović, D. 682
Folland, G.B. 153
Föllmer, H. 111, 317
Fouque, J.-P. 330, 351
Frachot, A. 690
Friz, P. 338, 379
Fukasawa, M. 971, 1049

Galton, F. 121
Gao, M. 355
Garman, M.B. 590
Gatarek, D. 4, 687
Gatheral, J. 338, 351, 356, 379, 382
Gatto, M.A. 409, 414, 453, 463
Geman, H. 486, 490, 576, 598, 690, 1120
Gerber, H.U. 566, 807
Glasserman, P. 809
Goldman, M.B. 409, 414, 453, 463
Gradshteyn, I.S. 341, 1061
Gram, J.P. 790
Gray, P. 1185
Greene, W.H. 620
Gueye, D. 377
Guirreri, S. 663

Hagan, P.S. 330, 357, 387, 1058
Han, J. 355
Handley, J.C. 511
Hardy, G.H. 518
Harrison, J.M. 73, 74, 207, 210
Haug, E.G. 1106
Heath, D. 4, 674
Hefter, M. 190
Henry-Labordère, P. 387
Herzwurm, A. 190
Heston, S.L. 315, 330, 347, 348, 357
Heynen, R.C. 1175
Hiriart-Urruty, J.-B. 33
Hirsch, F. 163
Ho, S.Y. 618
Hobson, D.G. 315
Huang, B. 644
Hughston, L.P. 645
Hull, J. 357, 618

Ikeda, N. 170, 276
Ingersoll, J.E. 264, 307, 614
Itô, K. 3

Jacka, S.D. 557
Jacod, J. 819
Jacquier, A. 317, 1034
Jaillet, P. 556
Jain, A. 358
Jamshidian, F. 598, 699, 720
Jarrow, R. 4, 674
Jarrow, R.A. 315
Jeanblanc, M. 463, 805
Jensen, J. 131, 482, 518, 553, 706
Joshi, M.S. 604

Kakushadze, Z. 643
Kakutani, S. 390
Kallenberg, O. 852
Kamal, M. 359
Kani, I. 374
Karolyi, G.A. 644
Kat, H.M. 1175
Kemna, A.G.Z. 483
Keynes, J.M. 610
Kim, Y.-J. 701
Klebaner, F. 314, 1023, 1134
Kloeden, P.E. 187
Kohlhagen, S.W. 590
Kopp, P.E. 545, 556
Korn, E. 809
Korn, R. 809
Kreps, D.M. 73, 74
Kroisandt, G. 809
Kumar, D. 330, 387, 1058

Lacombe, G. 163
Lamberton, D. 111, 500, 556
Lapeyre, B. 111, 500, 556
Lawi, S. 330, 614
Lee, R. 334, 359
Lee, S.B. 618
Lemaréchal, C. 33
Lesniewski, A. 357
Lesniewski, A.S. 330, 387, 1058
Leukert, P. 317
Leung, T. 320
Levy, E. 491
Lévy, P. 193
Li, Y. 355
Liinev, J. 497
Lim, J.W. 477
Lindström, E. 1186
Ling, C.-T. 511
Lipton, A. 225
Littlewood, J.E. 518
Litzenberger, R.h. 373
Loewenstein, M. 315

Longstaff, F.A. 558, 560, 644
Lyashenko, A. 678
Lyuu, Y.-D. 423

Maire, B. 971, 1049
Mamon, R.S. 1204
Margrabe, W. 596
Maringer, D.G. 1068
Marsh, T.A. 617, 646
Martini, C. 355
Matsumoto, H. 487
Mayer, P. A. 348
Meier, D.M. 645
Melnikov, A.V. 317, 907
Menkens, O. 511
Mercurio, F. 341, 630, 686, 716, 1193
Merton, R.C. 4, 597, 796
Meyer, P.A. 535
Mikosch, T. 949
Milevsky, M.A. 497
Milne, J.S. 1262
Mörters, P. 391
Morton, A. 4, 674
Musiela, M. 4, 673, 687

Nechaev, M.L. 317
Nelson, C.R. 679
Neuberger, A. 359
Nguyen, H.T. 737
Nikodym, O.M. 280
Norris, J.R. 741
Novikov, A. 282

Øksendal, B. 111, 293, 761, 805

Paley, R. 158
Palmer, K.J. 511
Papanicolaou, A. 330, 379
Papanicolaou, G. 330, 351
Peng, S. 311
Peres, Y. 391
Persson, J. 65, 425
Petrachenko, Y.G. 907
Pintoux, C. 638, 639
Pironneau, O. 377
Pitman, J. 819
Platen, E. 187
Pliska, S.R. 207, 210
Poisson, S.D. 735
Pólya, G. 518
Prayoga, A. 333, 641
Profeta, C. 404
Proske, F. 111, 293, 805

Protter, P. 180, 186, 282, 295, 315, 582, 595, 599, 628, 629, 819, 1029

Radon, J. 280
Rebonato, R. 331
Reiner, E. 435, 438
Revuz, D. 153, 1030
Rochet, J.-C. 576, 598
Rogers, C. 502
Rosenfeld, E.R. 617, 646
Ross, S. 48, 84
Ross, S.A. 75, 264, 307, 614
Rouah, F.D. 348
Rouault, A. 488
Roynette, B. 404
Rubinstein, M. 75, 435, 438
Rudebusch, G.D. 682
Rudin, W. 163
Ruiz de Chávez, J. 111
Rutkowski, M. 668
Ryzhik, I.M. 341, 1061

Samuelson, P.A. 3
Sanders, A.B. 644
Santa-Clara, P. 694
Sato, K. 775
Schied, A. 30, 36, 73, 74, 111, 123
Schoenmakers, J. 715, 716
Scholes, M. 3, 4, 210, 231
Schoutens, W. 348
Schröder, M. 489, 490
Schroeder, T.C. 9
Schwartz, E.S. 558, 560
Scorsese, M. 364
Sharpe, W.F. 75
She, Q.H. 355
Shi, Z. 502
Shimbo, K. 315
Shiryaev, A.N. 207, 210
Shiu, E.S.W. 566, 807
Shreve, S. 404, 413, 432, 506, 541, 562, 607, 1083
Siegel, A.F. 679
Sircar, K.R. 320, 330, 351, 379
Sølna, K. 330, 351
Sornette, D. 694
Sosin, H.B. 409, 414, 453, 463
Stähli 1068
Steele, J.M. 554
Stroock, D.W. 851
Sulem, A. 761
Svensson, L.E.O. 679

Tagasovska, N. 377

Tanaka, K. 790
Tankov, P. 761, 768, 774, 783, 805
Tatlow, G. 272
Teng, T.-R. 598, 704, 715
Thales 5
Thiele, T.N. 789
Tistaert, J. 348
Toy, B. 651
Turnbull, S.M. 491

UCLES, MOE & 827
Üstünel, A.S. 556
Uy, W.I. 639

Vanmaele, M. 497
Vašíček, O. 4, 610, 630
Vatter, T. 377
Večeř, J. 506
Vidal-Tomás, D. 229
Volkov, S.N. 317
Vorst, A.C.F. 483
Vorst, T. 907

Wakeman, L. 491
Wang, Y. 229
Watanabe, S. 170, 276
Watanabe, T. 790
Wei, X. 716
Whaley, R.E. 559, 561
White, A. 357, 618
Widder, D.V. 256
Wiener, N. 2, 158
Willard, G.A. 315
Williams, D. 111
Wilmott, P. 565
Wong, H.Y. 65, 425
Woodward, D.E. 330, 387, 1058
Wu, X. 653
Wunsch, M. 971, 1049

Yamada, T. 790
Yang, Z. 511
Yor, M. 153, 330, 404, 486–488, 490, 638, 1030, 1120
Yu, J.D. 494, 640

Zhang, P.G. 1106
Zhang, Q. 355
Zou, J. 359

This text is an introduction to the pricing and hedging of financial derivatives, including vanilla and exotic options, by stochastic calculus and partial differential equation methods. The presentation is done both in discrete and continuous-time financial models, with an emphasis on the complementarity between algebraic and probabilistic methods. In particular it covers the pricing of some interest rate derivatives, of American options, of exotic options such as barrier, lookback and Asian options, and stochastic models with compound Poisson jumps. The text is accompanied with a number of figures and simulations, and includes numerous examples based on actual market data. The concepts presented are also illustrated by 389 figures, and 290 exercises and 18 problems with complete solutions. It also includes 45 Python codes and 104 R scripts for illustrations based on market data.

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